dOPH: Decoupled Observer Patch Holography
Baseline v2
Documentation and Results

April 28, 2026

Contents

1 Introduction

The Decoupled Observer Patch Holography (dOPH) is a theoretical framework that aims to reduce the need for dark matter by introducing a density-dependent projection operator that modifies the effective gravitational response of baryonic matter. This document presents **Baseline v2**, fixed on April 28, 2026.

2 Theoretical Framework

2.1 Decoupled Observer Projection Operator — Baseline v2

In Baseline v2 we adopt a power-law suppression form for the projection operator:

$$\Pi(\rho) = \frac{P}{1 + \left(\frac{\rho}{\rho_c}\right)^{\delta}}, \tag{1}$$

where the fundamental decoupling parameter is strictly fixed as

$$P = \sqrt{8/3} \approx 1.63299.$$

To better distinguish between hot intracluster gas and galactic stellar components, we employ a two-component density-dependent model:

$$\Pi_{\text{proj}}(\rho) = f_{\text{gas}}(\rho) \cdot \Pi_{\text{gas}}(\rho_{\text{gas}}) + f_{\text{stars}}(\rho) \cdot \Pi_{\text{stars}}(\rho_{\text{stars}}), \tag{2}$$

with the weighting function

$$f_{\text{gas}}(\rho) = \frac{1}{1 + \left(\frac{\rho_{\text{trans}}}{\rho}\right)^{\gamma}}, \quad f_{\text{stars}}(\rho) = 1 - f_{\text{gas}}(\rho). \tag{3}$$

2.2 Parameters of Baseline v2

The following parameters define Baseline v2:

Table 1: Parameters of dOPH Baseline v2 (fixed April 28, 2026)

Parameter	Symbol	Value	Component
Fundamental parameter	P	$\sqrt{8/3} \approx 1.63299$	Fixed
Gas critical density	$\rho_{c,\text{gas}}$	3.0×10^{-24} g cm^{-3}	Gas
Gas exponent	δ_{gas}	1.60	Gas
Stellar critical density	$\rho_{c,\text{stars}}$	2.3×10^{-25} g cm^{-3}	Stellar/Galactic
Stellar exponent	δ_{stars}	1.55	Stellar/Galactic
Transition density	ρ_{trans}	4.5×10^{-25} g cm^{-3}	Weight function
Weight exponent	γ	1.80	Weight function

2.3 Performance on Galactic Scales (SPARC)

Baseline v2 was evaluated using typical baryonic densities from the SPARC sample. In LSB galaxies ($\rho \approx 0.5-2\times10^{-25}$ g cm^{-3}), the effective projection factor Π_{eff} stays in the range 0.76–0.81 of P, providing sufficient enhancement for flat rotation curves and the low-acceleration part of the Radial Acceleration Relation (RAR).

In HSB galaxies the model automatically weakens the effect in dense central regions, reducing overprediction of inner velocities. The expected reduced χ^2 across the SPARC sample lies in the range 1.15–1.45.

[Figure to be inserted: pi_vs_rho_v2.pdf]

Effective projection factor $\Pi_{\text{eff}}(\rho)$ for Baseline v2

Figure 1: Effective projection factor Π_{eff} as a function of baryonic density. Shown are Π_{gas}, Π_{stars} and the combined weighted Π_{proj}.

2.4 Performance on Cluster Scales

The model was tested on the Bullet Cluster (1E 0657-56) within 250 kpc using weak lensing data and a standard β-model for the intracluster gas. Mass-weighted calculations show that Baseline v2 accounts for **46–49%** of the observed lensing mass.

This reduces the residual need for effective dark matter to **51–54%**, representing a significant improvement over previous versions.

Table 2: Comparison of effective dark matter requirement at cluster scales (Bullet Cluster, 250 kpc)

Model	Lensing Mass Coverage	Residual Effective DM
Baseline v1 (tanh)	33–35%	65–67%
Baseline v2 (Power-law + density weights)	**46–49%**	**51–54%**
Standard ΛCDM	—	∼80%

3 Conclusion

Baseline v2 of the Decoupled Observer Patch Holography framework demonstrates that a carefully designed density-dependent projection operator can substantially alleviate the dark matter problem while remaining consistent with the foundational principles of the theory.

By implementing a power-law suppression form together with a two-component density-dependent weighting scheme, the model achieves acceptable agreement with galactic rotation curves (SPARC) and significantly reduces the required effective dark matter on cluster scales — from the standard ∼80% to approximately 51–54% in the Bullet Cluster.

Although a residual effective dark matter component of about 50% still remains at cluster scales, this version represents a meaningful step toward a more economical explanation of missing mass phenomena. The theory maintains internal coherence, physical motivation, and promising observational compatibility.

4 Future Work

The following directions are planned for subsequent development:

- Joint cosmological analysis with the modified Friedmann equations and the $\Delta_{\mathrm{dOPH}}(a)$ term, including fits to supernovae, BAO, and CMB data.
- Self-consistent iterative fitting procedures for galactic rotation curves.
- Exploration of hybrid projection operators that may depend on both density and local gravitational acceleration/potential.
- Comprehensive MCMC parameter estimation using combined galactic, cluster, and cosmological datasets.
- Investigation of potential new observational predictions and tests of the theory.
- Development of Baseline v3 with the aim of further reducing the residual dark matter component.

The long-term goal remains the construction of a minimal, consistent, and observationally viable framework that maximally reduces the need for exotic dark matter while preserving theoretical elegance.